

Demo: Coupled-Coil Showpieces

ChatGPT edition — two safe finales for
the course instrument

Visible low-voltage effects without exposed high voltage

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

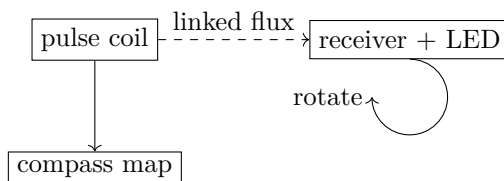
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- Completed coupled-coil instrument
- LED with 1 k Ω resistor
- Small compass
- Graph paper

Predict first

At what receiver orientation should the induced indication be weakest?



Investigate

1. **Orientation null:** pulse identically while rotating the receiver in 15-degree steps. Record LED response or meter peak and find the minimum.
2. **Field portrait:** remove the receiver; use the compass to map primary field direction at grid points during brief, consistent current pulses.
3. Overlay both maps. Ask whether the weak coupling orientation matches little linked flux.
4. Power off and disconnect before every repositioning.

Explain from the evidence

These effects are the legitimate finale of the learner course: field direction, changing flux, geometry, and energy transfer all become visible without sparks. An LED is directional and nonlinear, so meter data are better for comparison; the LED supplies immediacy, not precision.

Change one thing

Predict and test what a shared core changes. Keep pulse duration, source, and coil geometry fixed and report trial-to-trial spread.

Evidence record

Prediction

One change

Observation

Claim + because

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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